PROJECT REPORT

On

Driver Drowsiness Detection System using OpenCV

Submitted in Partial Fulfillment of Award of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

BY

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

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CERTIFICATE

This is to certify that the project work entitled Driver Drowsiness Detection System submitted by Ayush Chauhan 241121060069 Anisha Padhi 2411060093 Akshata Kuar 2411021060107 Ayesha Ali 2411021060159 in partial fulfilment for the award of the degree of Bachelor of Technology (Computer Science Engineering) of Alliance University, is a Bonafide work accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr R C Karpagalakshmi
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DECLARATION

I/We hereby declare that the project entitled "Driver Drowsiness Detection System Using Open CV" submitted by me/us in the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (CSE) of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of Dr R C Karpagalakshmi

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

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TABLE OF CONTENTS

1. INTRODUCTION

2. LITERATURE SURVEY

3. SYSTEM DESIGN

4. SYSTEM IMPLEMENTATION

5. RESULTS AND DISCUSSION

6. TESTING

7. CONCLUSION AND FUTURE ENHANCEMENTS

8. RFERENCES & BIBLIOGRAPHY

1. INTRODUCTION

Road accidents caused by driver fatigue and drowsiness are a major concern worldwide. Long hours of driving, lack of sleep, and monotonous road conditions significantly reduce driver alertness, leading to dangerous situations. According to global statistics, a considerable percentage of road accidents occur due to driver drowsiness.

To address this issue, the **Driver Drowsiness Detection System** has been developed. This system uses computer vision techniques to monitor the driver's facial features in real-time and detect signs of fatigue such as prolonged eye closure and yawning.

The system is implemented using **Python and OpenCV**, leveraging Haar Cascade classifiers for detecting facial features like eyes, face, and mouth. Based on these detections, the system determines whether the driver is alert, in a warning state, or drowsy.

The goal of this project is to create a **real-time, efficient, and low-cost solution** that enhances road safety by alerting drivers before a critical situation arises.

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2. LITERATURE SURVEY

Various approaches have been proposed for detecting driver drowsiness:

2.1 Image Processing-Based Methods

These methods use cameras to track facial features such as:

- Eye blinking rate
- Eye closure duration
- Mouth opening (yawning)

Advantages:

- Non-intrusive
- Cost-effective

Limitations:

- Sensitive to lighting conditions
- May fail with occlusions (e.g., sunglasses)

2.2 Sensor-Based Methods

Use physiological signals like:

- EEG (brain signals)

- Heart rate monitoring

Advantages:

- High accuracy

Limitations:

- Expensive
- Intrusive and uncomfortable for drivers

2.3 Machine Learning Approaches

- Deep learning models like CNNs
- Real-time classification of alert vs drowsy

Advantages:

- High accuracy

Limitations:

- Requires large datasets
- High computational cost

2.4 Proposed System

The current project uses:

- Haar Cascade Classifiers (lightweight)
- Real-time video processing
- Rule-based logic for detection

This makes it:

- Fast
- Easy to implement
- Suitable for real-time applications

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3. SYSTEM DESIGN

3.1 System Architecture

The system consists of the following modules:

1. **Video Capture Module**
    - Captures real-time video from webcam

2. **Face Detection Module**
    - Detects faces using Haar Cascade
3. **Eye Detection Module**
    - Detects eyes in the upper half of the face
4. **Mouth Detection Module**
    - Detects yawning in the lower half
5. **Drowsiness Detection Logic**
    - Uses counters and thresholds
6. **Alert System**
    - Displays warnings on screen

3.2 Workflow

1. Start camera
2. Capture frame
3. Convert to grayscale
4. Detect face
5. Detect eyes and mouth
6. Apply logic:
    - Eyes closed → increase counter
    - Yawning → increase counter
7. Determine status:
    - Alert
    - Warning
    - Drowsy
8. Display output

3.3 Algorithms Used

- Haar Cascade Classifier
- Frame-based counting logic
- Threshold-based classification

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4. SYSTEM IMPLEMENTATION

4.1 Technologies Used

- **Programming Language:** Python
- **Libraries:**
    - OpenCV
    - NumPy
    - Time

4.2 Key Components of Code

Class: DrowsinessDetector

This class handles all detection logic.

4.2.1 Initialization

- Loads Haar Cascade files:
  - Face
  - Eyes
  - Mouth
- Initializes:
  - Eye counter
  - Yawn counter
  - Score

4.2.2 Detection Function

def detect(self, gray):

- Detects face using detectMultiScale
- Splits face into:
  - Upper half → eyes
  - Lower half → mouth
- Returns detection results

4.2.3 Update Function

def update(self, results):

Logic:

- If 2 eyes detected → reset counter
- If 1 eye → partial increment
- If no eyes → increase counter

Drowsiness condition:

```
if eye_counter >= 20:
    drowsy = True
```

Yawning:

```
if yawn_counter >= 15:
    yawning = True
```

4.2.4 Drawing Function

def draw(self, frame, results, drowsy, yawning, score):

- Draws bounding boxes:
  - Green → normal
  - Orange → warning
  - Red → drowsy
- Displays:
  - Score

- Status
- Alert message

4.2.5 Main Function

- Starts webcam
- Reads frames continuously
- Applies detection and update
- Displays output window

---

5. RESULTS AND DISCUSSION

5.1 Observations

Condition	Output
Eyes open	ALERT
Partial closure	WARNING
Eyes closed (long)	DROWSY
Yawning detected	ALERT triggered

5.2 Output Features

- Real-time face tracking
- Eye and mouth detection
- Status display:
  - ALERT
  - WARNING
  - DROWSY
- Visual alert: **"WAKE UP!"**

5.3 Performance

- Works in real-time
- Accurate under normal lighting
- Slight delay due to processing

5.4 Limitations

- Affected by poor lighting
- Cannot detect through sunglasses
- Haar cascades are less accurate than deep learning

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6. TESTING

6.1 Test Cases

Test Case	Input	Expected Output	Result
Normal driving	Eyes open	ALERT	PASS
Slow blinking	Partial closure	WARNING	PASS
Eyes closed	Long closure	DROWSY	PASS
Yawning	Mouth open	ALERT	PASS

6.2 Testing Environment

- OS: Windows
- Python Version: 3.x
- Webcam: Laptop camera

6.3 Error Handling

- Camera not opening:
  - Displays error message
- No face detected:
  - System continues safely

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7. CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The Driver Drowsiness Detection System successfully detects fatigue indicators using computer vision techniques. The system provides real-time alerts and helps prevent accidents caused by drowsiness.

It is:

- Efficient
- Cost-effective
- Easy to deploy

7.2 Future Enhancements

1. **Use Deep Learning Models**
   - CNN for higher accuracy
2. **Audio Alerts**
   - Add alarm sound
3. **Mobile Integration**

    - Android app support
4. **Infrared Camera**
    - Night detection improvement
5. **Head Pose Detection**
    - Detect nodding
6. **Cloud Integration**
    - Data logging and monitoring

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